

1	Supply of Relay rack universal type as per drg.no.S&T/MFT/291 with
2	Supply of Cable termination racks size 880mm x 2 Meters
3	Supply of terminal block 1 way PBT 60mm as per RDSO Drg.No.
4	Supply of terminal block 6 way PBT 25mm as per RDSO Drg.No.
5	Supply of Non-deteriorating type of fuse holders 20A capacity - for Railway signalling
6	Supply of Non-deteriorating type of fuse links 2 Amps- electric
7	Supply of Non-deteriorating type of fuse links 16 Amps- electric
8	Supply of Non-deteriorating type of fuse links 10 Amps- electric
9	Supply of Non-deteriorating type of fuse links 6 Amps- electric
10	Final location survey of cableroute and final preparation of cable route plan as instructions of site incharge.
11	Breaking of metal/ tar road and refilling after laying the cables as per drawing no.SDO/CABLE LAYING/010. The surface of the road shall be concreted to match with the old surface
12	Excavation of square coil pit upto a depth of 1 Mtr. X 1.5 Mtr. For providing coil at location box, point, signals etc., This includes back filling and ramming of trench / pit after laying the cable and consolidation of soil.
13	Making a simulation test board (6 road means 2 Nos. 3 road stations) (Unit : Per Road)
14	Cutting of rock to a depth of 200mm and width of 200 mm where rock is visible from surface to a depth more than 300 mm up to 550mm, excavation of trench with width 300mm above rock concreting 200mmx200mm after laying of cable with 1 cement: 2 sand : 4 graded stone aggregate 20mm nominal size and curing of concrete to the required number of days, refilling with excavated soil ramming and consolidation except in bridges, culverts, level crossings, track crossings, cable pits, rocky soil including marking of cable alignment, clearing of debris etc., complete to the finished item of work as directed by site in-charge. Drg.No.SK/CN/NEW/3. All the material is to be arranged by contractor
15	When rock is visible for more than 0.5 metre and up to 0.8 metre from ground level, excavating the trench minimum 0.5 metre and maximum 0.8 metre, laying 6-quadrant cable as per the details shown in drawing no 3 and concreting 200mmx200mm over the rocky surface after leveling with soil/sand. (Chase cutting not required).
16	Supply & Laying of G.I Pipes 100mm dia. medium quality to specification IS.1239 (part-I) 1990 and galvanised to specification IS.4736-1986 On bridges, culverts, trench with channel ramps on either end of the culvert/bridges. Clamping arrangement with MS brackets for fixing of GI pipes on bridges/culverts, small holes of 5mm are to be drilled on the pipe before laying of the pipes.

17	Foundations, erection, fixing of 'E' type locks, painting the box and painting of termination & other particulars of the Full location box. Metal Size of the aggregate should not be more than 40mm. Work to be done as per prescribed drawing no. SK/CN/1/89-A & B- All the material is to be arranged by the contractor except E- type lock (E-type lock will be supplied by RIYS)
18	Excavation of earth and providing earth pipe of size 50mm dia, 3.5mm thick 2.0 m. long G.I. pipe with masonry tub constructed around at earth locations as specified by site Engineer. Alternate layers of 30kgs- chrcoal and 30kgs-common salt to be filled in the pit while closing. The pipe perforation etc., to be carried out as per drawing No. G/SG/09/2002. The earth pipe to be connected to the function with cable as supplied by Railways. The reading should be painted on the masonry tub. The masonry tub of 300X300X300mm should be covered with cement slab. (All materials to be supplied by the contractor except cable.)
19	Test lamp arrangement in LBs/RE cuttings. (Auto mobile handle type) (Unit ; Per Location)
20	Straight through / derivation with/without transformer joints: Installation Thermo shrinkable jointing kits for straight through/ derivation joints for LC gate, pump houses etc where VF transformers are not required and derivation joint by using 2T transformer (to be housed in the joint) . This includes tapping of nominated Quad at a specific location shown by the Railway Engineer at site to the jelly field cable already laid. All material including valco tapes except thermo shrinkable jointing kit shall be supplied by the contractor. (Refer Annexure for instructions on cable jointing). (Unit ; Joint)
21	Provision and laying in position cement concrete of 1 cement:3 sand:6 graded stone aggregate of 20mm normal size excluding cost of cement, centering and shuttering (including transportation of all the material) .
22	Cement 53 grade (@ the Rate (in Rupees) of 2.2 quintals per Cum of concrete including transportation
23	Transportation of materials from any Stores to Site and vice-versa including loading and unloading. -Lead above 50 Kms. (Unit : PTPK)
24	Excavation of trench to a depth of 1200mm and width of 300mm at bottom in hard morum soil as per RDSO Sketch no SDO/CABLE LAYING /003 including marking of cable alignment, clearing of jungle, bushes, trial pits if required, refilling with excavated soil, ramming and consolidation after laying of cables complete to the finished item of work as directed by supervisor in-charge.
25	Supply of Manageable L2 switch 12-Port 10/100/1000 POE+BaseT with 2 Uplink optical 1G Ports upgradable to 2x10G SFP+uplink ports similar to RUCKUS ICX7150 C12P or better including all accessories licence, software up gradation and as per specification given in Annexure - XVIII

26	Supply of Puncom make 2Mbloop protected programmableprimary digital drop/ insertmultiplexure with conferencefacility as per RDSO spec No.IRS-TC 68-2012 With Amend. 1or latest, wired for 30 circuitscomplete with power supply,control and basic /commoncards (ie PSU-2, TME-1, LPC-1)remote control supervisionfacility and all other accessoriesetc
27	Supply of 2W/4W DTMF Headquarter control room Equipmentas per RDSO specNo.IRS-TC60/2007with amendment No.1latest. (To be procured fromRDSO approved firms)
28	Supply of 10/100/1000 (1GBPS) single mode mediaconverter with accessoriesincluding connecting cables.Make: MRO-TEK/D-Link/RAD/Team Engineer/Digisol orbetter as specified by EngineerIncharge.
29	Supply of Amplifier 80W Ahujamake (SSB-80M) working 230VAC/12V DC
30	Supply of printer Dot matrixcompatable to existing datalogger. Printer Dot matrix 80column, 24 pin, 300 CPSequivalent to TVS-MSP 450champion or better
31	Supply of Voltage Stabilizer,capacity: 3 KVA, 1 Ph; InputVoltage 140-270V AC; Outputvoltage 230V (+/-) 10%, with adouble pole switch of 32 Amps
32	Laying of PVC switch boardcables /CAT6/ power /OFC6F/12F cables in the PVCcasing/PVC pipes as per theinstruction of site incharge.
33	Supply And Installation of 42U x800mm x 1000 mm with frontperforated door and rear dualperforated door, side panels,5/15 AMPs Vertical PDU -2 Nos,4 X Fans, 2 x Shelf with earthingstrip (copper)-1 with fasteners-2pac, rack mountable DCDB panelwith one common copper stripand 4 MCB -6A & 2MCB-16A. asper specification given inAnnexure - XVIII
34	Installation Labour, Design andProject Management,Programming, Operation andMaintenance Manuals - includingEquipment Manuals & Training - Operation and Maintenance.Provision of ring earth with min03 MFE earths & below 1 ohmearthing and connectivity toEquipment
35	Supply and laying of CAT-6 cablepatch cards-2mtrs length withRJ-45 Jacks on both ends Make:D-Link/Bestnet
36	Supply and installation of 5KVAUPS with battery backup as perthe specification given inAnnexure - XVIII
37	Supply & installation of 2KVAUPS systems with battery backup of 120 mins Make:APC/Numeric/Emerson withBattery minimum 65 AH each
40	Supply and installation of Wi-Fiaccess point with POE facilityMake: TP link/D-Link.
41	Fabrication and Supply ofPowder coated Aluminum ladderof Size 38mmX38mmX5mm and 250 mm wide , with 38mmX5mm flat fixed at intervals ofnot more than 250 mm. All nuts,bolts, PVC sleeves, insulatorsetc. are to be arranged by thecontractor

44	Releasing of relay rackscomplete with its wiring, relaybase, supporting frames, cableladder, indoor/outdoor cablesetc. Axle counter wiring, Datalogger and its wiring, panel andits wiring completely and any ofthe miscellaneous wiring inrelay/equipment/panel room etc.complete.
45	Supply and binding of Albumtype booklet with C- size plastictransparent covers of approvedquality
46	Supply and installation of wagoconnectors FTB set (per unitmeans per location)
47	Supply and installation of wagoconnectors UFTB set (Un-fusedterminal block) (Unit : Perlocation)
48	Supply and installation of wagoconnector End Plate set (End Clamp) (Unit ; Per Location)
49	Supply and installation of DinRail (Unit : Per Location)
50	Supply, installation, testing andcommissioning of 24 port Gigabitwith 2 nos of loaded 1G SFPmodules PoE+ switch(L-2Switch) with 370W or higher